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<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Calculus in Animal Husbandry: Kedah-Kelantan Cattle Growth
Rates</title>
  <!-- Tailwind CSS for modern UI styling -->
  <script
src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
  <!-- Chart.js for smooth mathematical animations -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    @import
url('https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@4
00;500;600;700&display=swap');
    body { font-family: 'Plus Jakarta Sans', sans-serif; }
  </style>
</head>
<body class="bg-slate-50 text-slate-800 min-h-screen py-8 px-4 sm:px-6
lg:px-8">

  <div class="max-w-5xl mx-auto">
    <!-- Header Banner -->
    <header class="bg-gradient-to-r from-emerald-700 to-teal-600
rounded-2xl p-6 md:p-8 text-white shadow-lg mb-8">
      <span class="bg-emerald-500/30 text-emerald-100 text-xs
font-semibold uppercase tracking-wider px-3 py-1 rounded-full
backdrop-blur-xs">
        SFH: Animal Husbandry Technology Application
      </span>
      <h1 class="text-2xl md:text-4xl font-bold mt-3
tracking-tight">Applying Derivatives to Livestock Production</h1>
      <p class="text-emerald-100 mt-2 max-w-2xl text-sm
md:text-base">
        Using Calculus to optimize feeding schedules, detect
health anomalies, and pinpoint the exact day of peak growth
performance in native Malaysian beef cattle.
      </p>
    </header>

    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">
      <!-- Left Side: Mathematical Formulas & Biological
Explanation -->
      <div class="lg:col-span-1 space-y-6">
        <!-- Scenario Context Card -->

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</span> Biological Scenario
  </h2>
  <p class="text-xs text-slate-600 leading-relaxed
mb-3">
    In Malaysian beef production, managing
    **Kedah-Kelantan (KK) cattle** efficiently requires knowing *when*
    they grow the fastest. Tracking cumulative weight  $f(x)$  is good, but
    calculating the **derivative  $f'(x)$ ** reveals the exact growth
    velocity.
  </p>
  <p class="text-xs text-slate-600 leading-relaxed">
    When the growth rate drops significantly, it
    signals the animal has hit biological maturity—meaning continuing to
    feed them high-cost concentrates yields diminishing returns.
  </p>
</div>

<!-- Mathematical Formulas Card -->
</span> The Mathematics
  </h2>

  <div class="space-y-4 text-xs">
    <div>
      <span class="block font-semibold
text-slate-700 mb-1">Growth Curve  $f(x)$  [Gompertz Model]:</span>
      <div class="bg-slate-50 p-2.5 rounded
font-mono text-emerald-700 font-bold text-center border
border-slate-200">
        
$$f(x) = A \cdot e^{(-b \cdot e^{(-k \cdot x)})}$$

      </div>
    </div>
    <div>
      <span class="block font-semibold
text-slate-700 mb-1">Derivative  $f'(x)$  [Chain Rule Applied]:</span>
      <div class="bg-slate-50 p-2.5 rounded
font-mono text-teal-700 font-bold text-center border
border-slate-200">
        
$$f'(x) = f(x) \cdot b \cdot k \cdot e^{(-k \cdot x)}$$

      </div>
    </div>
  </div>

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        </div>

        <div class="text-slate-500 space-y-1 pt-2
border-t border-slate-100">
            <p><strong>x:</strong> Animal age in
weeks</p>
            <p><strong>A (Asymptotic Weight):</strong>
320 kg (Mature mass)</p>
            <p><strong>b (Biological scaling
constant):</strong> 2.85</p>
            <p><strong>k (Growth rate
modifier):</strong> 0.042</p>
        </div>
    </div>
</div>

    <!-- Reference Data Verification -->
    <div class="bg-slate-100 p-4 rounded-xl border
border-slate-200 text-xxs text-slate-500">
        <span class="font-bold text-slate-700 block
mb-1">📌 Reference Data Baseline:</span>
        Parameters calibrated against growth performance
parameters published in local literature, including MARDI (Malaysian
Agricultural Research and Development Institute) and the Malaysian
Journal of Animal Science (MJAS) historical reports on *Kedah-Kelantan
(KK) breed performance profiles*.
    </div>
</div>

    <!-- Right Side: Live Chart and Dynamic Interactive
Readout -->
    <div class="lg:col-span-2 space-y-6">
        <!-- Dynamic Calculus Readout Box (changes on click)
-->
        <div id="readout-box" class="bg-amber-50 border
border-amber-200 p-5 rounded-xl shadow-xs transition-all
duration-300">
            <div class="flex items-start gap-4">
                <div class="bg-amber-500 text-white p-3
rounded-lg text-xl font-bold" id="readout-week">
                    W40
                </div>
                <div class="flex-1">
                    <h3 class="font-bold text-amber-900
text-md" id="readout-title">
                        Dynamic Derivative Engine Ready
                    </h3>
                    <p class="text-xs text-amber-800 mt-1

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leading-relaxed" id="readout-body">
    Click anywhere on the chart grid
points below to calculate the exact instantaneous rate of weight gain
($f'(x)$) for that specific week of the animal's life cycle.
    </p>
    </div>
    </div>
    </div>

    <!-- Chart Container Canvas -->
    <div class="bg-white p-4 md:p-6 rounded-xl shadow-xs
border border-slate-100">
        <div class="flex flex-col sm:flex-row
justify-between items-start sm:items-center gap-2 mb-4">
            <div>
                <h3 class="text-base font-bold
text-slate-950">Kedah-Kelantan Body Mass Trajectory</h3>
                <p class="text-xs text-slate-500">Click
individual dots to analyze biological calculus variables.</p>
            </div>
            <div class="flex gap-4 text-xs font-medium">
                <span class="flex items-center
gap-1.5"><span class="w-3 h-3 bg-emerald-600 rounded-full
inline-block"></span> Total Weight ($f(x)$)</span>
                <span class="flex items-center
gap-1.5"><span class="w-3 h-3 bg-teal-500 rounded-full
inline-block"></span> Growth Velocity ($f'(x)$)</span>
            </div>
        </div>

        <div class="relative w-full h-[320px]
md:h-[400px]">
            <canvas id="growthChart"></canvas>
        </div>
    </div>
</div>

    <footer class="mt-12 text-center text-xs text-slate-400
border-t border-slate-200 pt-4">
        Animal Husbandry Technology Simulator • Powered by
Calculus and HTML5 Canvas Engine
    </footer>
</div>

    <!-- Application Controller Logic -->
    <script>
        // Parameters for typical Malaysian Kedah-Kelantan Cattle

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Growth Model

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const A = 320;    // Asymptotic mature weight (kg)
const b = 2.85;   // Initial growth displacement parameter
const k = 0.042;  // Growth scaling constant per week

// Generate Dataset values from week 0 to 120 (approx 2.3
years production timeline)
const weeks = [];
const weightData = [];
const derivativeData = [];

for (let x = 0; x <= 120; x += 4) {
    weeks.push(x);

    // Primary Gompertz function f(x)
    const fx = A * Math.exp(-b * Math.exp(-k * x));
    weightData.push(fx.toFixed(2));

    // Derivative function f'(x) -> returns change per week,
divided by 7 for daily rate
    const fPrimeX = fx * b * k * Math.exp(-k * x);
    const dailyRate = fPrimeX / 7;
    derivativeData.push(dailyRate.toFixed(3));
}

// Initialize Chart.js configuration
const ctx =
document.getElementById('growthChart').getContext('2d');
const growthChart = new Chart(ctx, {
    type: 'line',
    data: {
        labels: weeks,
        datasets: [
            {
                label: 'Total Weight f(x) (kg)',
                data: weightData,
                borderColor: '#059669',
                backgroundColor: 'rgba(5, 150, 105, 0.05)',
                borderWidth: 3,
                yAxisID: 'yWeight',
                tension: 0.3,
                pointRadius: 4,
                pointHoverRadius: 8,
                pointHoverBackgroundColor: 'ffffff',
                pointHoverBorderColor: '#059669',
                pointHoverBorderWidth: 3
            },
        ],
    },
    {
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        label: 'Growth Rate f\'(x) (kg/day)',
        data: derivativeData,
        borderColor: '#06b6d4',
        borderDash: [5, 5],
        backgroundColor: 'transparent',
        borderWidth: 2,
        yAxisID: 'yRate',
        tension: 0.3,
        pointRadius: 4,
        pointHoverRadius: 8,
        pointHoverBackgroundColor: 'ffffff',
        pointHoverBorderColor: '#06b6d4',
        pointHoverBorderWidth: 3
    }
    ],
    },
    options: {
        responsive: true,
        maintainAspectRatio: false,
        plugins: {
            legend: { display: false } // Custom legend layout
inside standard DOM text instead
        },
        scales: {
            x: {
                title: { display: true, text: 'Animal Age
(Weeks)', font: { size: 11, weight: '600' } },
                grid: { color: '#f1f5f9' }
            },
            yWeight: {
                type: 'linear',
                position: 'left',
                title: { display: true, text: 'Cumulative
Weight (kg)', font: { size: 11, weight: '600' } },
                min: 0,
                max: 350,
                grid: { color: '#f1f5f9' }
            },
            yRate: {
                type: 'linear',
                position: 'right',
                title: { display: true, text: 'Daily Gain
Derivative (kg/day)', font: { size: 11, weight: '600' } },
                min: 0,
                max: 0.8,
                grid: { drawOnChartArea: false } // Avoid grid
clashes
            }
        }
    }
}

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    },
    // Hook into interaction events to catch user clicks
    anywhere on plot points
    onClick: (event, activeElements) => {
      if (activeElements && activeElements.length > 0) {
        const index = activeElements[0].index;
        const weekValue = weeks[index];
        const currentWeight = weightData[index];
        const currentRate = derivativeData[index];

        updateReadout(weekValue, currentWeight,
currentRate);
      }
    }
  });

  // Function that executes dynamically when user clicks a value
point
  function updateReadout(week, weight, rate) {
    const readoutBox = document.getElementById('readout-box');
    const readoutWeek =
document.getElementById('readout-week');
    const readoutTitle =
document.getElementById('readout-title');
    const readoutBody =
document.getElementById('readout-body');

    // Transition UI element color schema to indicate live
analytical computation success
    readoutBox.className = "bg-emerald-50 border
border-emerald-200 p-5 rounded-xl shadow-xs transition-all
duration-300 transform scale-[1.01]";
    readoutWeek.className = "bg-emerald-600 text-white p-3
rounded-lg text-xl font-bold flex items-center justify-center
min-w-[70px]";

    readoutWeek.innerHTML = `W${week}`;
    readoutTitle.innerHTML = `Calculated Dynamics: Weight =
${weight} kg | f'(x) = ${rate} kg/day`;

    // Formulate custom layman bio-technological
interpretations depending on dynamic location along function curve
    let laymanExplanation = "";
    if (week <= 24) {
      laymanExplanation = `This is the accelerating growth
phase. The positive derivative tells us the animal is packing on body
mass faster and faster each day. Keep feeding high-protein rations

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right now to build structural frame bone and muscle tissue
efficiently.`;
    } else if (week > 24 && week <= 48) {
        laymanExplanation = `The animal is currently passing
or near its **Inflection Point** (approx week 25). At this point, the
derivative  $f'(x)$  peaks at its maximum height. The calf is gaining
weight faster right now than it ever will again. Feeding efficiency is
at maximum biological peak.`;
    } else if (week > 48 && week <= 84) {
        laymanExplanation = `Notice that while total weight is
still climbing, the growth rate derivative  $f'(x)$  is sloping
downward. This means the calf's growth velocity is slowing down. It is
entering the fattening stage, transitioning from building muscle to
laying down subcutaneous adipose fat layers.`;
    } else {
        laymanExplanation = `The derivative has dropped close
to zero ( $f'(x)$  \approx  $\text{\$}{rate}\text{\$}$  kg/day). The animal is at full
maturity, matching its ceiling asymptotic weight. Keeping this animal
in the feedlot past this week will cost you more money in Malaysian
palm kernel cake feed inputs than the tiny daily weight gains are
worth. Time to market!`;
    }

    readoutBody.innerHTML = laymanExplanation;

    // Clean transform bump highlight back down smoothly
    setTimeout(() => {
        readoutBox.classList.remove('scale-[1.01]');
    }, 150);
}
</script>
</body>
</html>

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